The period is

$$T=\frac{2\pi}{\omega}=2\pi\sqrt{\frac{(d^2+y_1^2)\,y_1}{(2y_1^2-d^2)\,g}}\,. \qquad (10)$$

4. P has gravitational energy and, from charge A, electrostatic potential energy. When P has coordinate y, its gravitational potential energy is

$$W_g=m_2gy\ ,$$

Gravity has zero potential at O; the electrostatic energy is

$$W_E=k\frac{qQ}{\sqrt{d^2+y^2}}\ .$$

The total potential energy is

$$W=W_g+W_E=m_2gy+k\frac{qQ}{\sqrt{d^2+y^2}}\ . \qquad (11)$$

At P's y, small y: electrostatic; large y: gravitational dominance. At P's stable equilibrium it is a minimum; at P's unstable equilibrium it is a maximum. Here $y=y_2$ is unstable for mass m_2. Thus at $y=y_2$ the potential has its maximum:

$$W(y_2)=W_{\max}=m_2gy_2+k\frac{qQ}{\sqrt{d^2+y_2^2}}\ . \qquad (12)$$

When P has coordinate y_3, its potential energy is

$$W(y_3)=m_2gy_3+k\frac{qQ}{\sqrt{d^2+y_3^2}}\ .$$

For $y_3<y_2$, any y_3 gives a fall to the bottom from rest. For $y_3>y_2$, release from rest reaches the tube bottom only if

$$W(y_3)>W(y_2)\ .$$

Thus y_3 must satisfy

$$y_3<y_2\ ; \qquad (13)$$

or

$$y_3>y_2\ \text{ and }\ m_2gy_3+k\frac{qQ}{\sqrt{d^2+y_3^2}}>m_2gy_2+k\frac{qQ}{\sqrt{d^2+y_2^2}}\ . \qquad (14)$$

Appendix: (1) can be written as